Ch. 25 HW #1 Solutions

Principles of Macroeconomics

February 2, 2026

Problems & Applications

2. (a) 2023:

$$P_{Cauliflower} = \frac{200}{100} = 2$$
$$P_{Broccoli} = \frac{75}{50} = 1.5$$
$$P_{Carrots} = \frac{50}{500} = 0.1$$

2024:

$$P_{Cauliflower} = \frac{225}{75} = 3$$
$$P_{Broccoli} = \frac{120}{80} = 1.5$$
$$P_{Carrots} = \frac{100}{500} = 0.2$$

(b) I'm going to assume that the typical consumption basket is 4 heads of Cauliflower, 2 bunches of broccoli, and 20 carrots. Your basket might be different. We aren't told what to assume, so you can assume whatever you want.

Year	Cost of Basket	CPI
2023	$4(2) + 2(1.5) + 20(0.1) = 13$	100
2024	$4(3) + 2(1.5) + 20(0.2) = 19$	$19/13 * 100 = 146$

(c) $\frac{146-100}{100} \cdot 100 = 46\%$

3. (a) Tennis Balls: 0
Golf Balls: $\frac{6-4}{4} \cdot 100 = 50\%$
Gatorade: $\frac{2-1}{1} \cdot 100 = 100\%$

(b) We're not told how to fix the basket, so it's up to you. I'm going to assume the typical person buys one tennis ball, one golf ball, and two bottles of gatorade.

Year	Cost of Basket	CPI	Inflation
2023	$1(2) + 1(4) + 2(1) = 8$	100	–
2024	$1(2) + 1(6) + 2(2) = 12$	$12/8 * 100 = 150$	$\frac{150-100}{100} \cdot 100 = 50\%$

(c) It should. There's been a quality improvement in Gatorade bottles, so the price increase is partially due to quality improvement and not just an increase in the cost of living. As a result, our inflation rate should be lower than the one we calculated in (b).

(d) It should. The introduction of new products increases the value of a dollar. That increased value offsets some of the increase in the cost of living, so our inflation rate should be lower than the one we calculated in (b).

4. You don't need me for this one.

5.